

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results_backup/dp_8_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,32.29933751890424,0.04900664415170795,0.07,0.007000000000000001,0.0020227477003257922,13.70424580230713,3.5219111457824708
2,32.88257406573976,,,,,13.735208779907227,3.522513412475586
3,33.490608923782005,,,,,13.773474458312988,3.52244154586792
4,34.03537829007678,,,,,13.8231083984375,3.5225142669677734
5,34.5790928287218,0.06690983654996545,0.09,0.011000000000000001,0.007239654355057395,13.886728021240234,3.522496696472168
6,35.31948951689446,,,,,13.964869668579102,3.5228670364379884
7,36.126836900261964,,,,,14.081374354553223,3.5267831939697265
8,36.630888192507676,,,,,14.17117515258789,3.5257721771240234
9,36.98664217515131,,,,,14.257159707641602,3.5239249969482422
10,37.696265293993996,0.03561013735332017,0.05,0.005,0.001212587264121489,14.396677098083495,3.529047723388672
FINAL,,0.03561013735332017,0.05,0.005,0.001212587264121489,14.396677098083495,3.529047723388672

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:58:54.374272",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.736760905458635,
      "test_metrics": {
        "mse": 13.68513099822998,
        "mae": 3.5215539653778074,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.73491219975885,
      "test_metrics": {
        "mse": 13.685111654663086,
        "mae": 3.5215516830444336,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ]
}
```

```
{
  "round": 3,
  "train_loss": 13.74112209662515,
  "test_metrics": {
    "mse": 13.685089517211914,
    "mae": 3.521549138259888,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.740059274786885,
  "test_metrics": {
    "mse": 13.685065866088868,
    "mae": 3.521546283340454,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.731428745961571,
  "test_metrics": {
    "mse": 13.685040350341797,
    "mae": 3.5215431350708006,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.726476204674901,
  "test_metrics": {
    "mse": 13.68501517791748,
    "mae": 3.5215401008605958,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.731320599997662,
  "test_metrics": {
    "mse": 13.684988584899902,
    "mae": 3.521536766433716,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.72987025365621,
  "test_metrics": {
```

```

        "mse": 13.684961358642578,
        "mae": 3.5215334129333495,
        "samples": 20000
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
{
    "round": 9,
    "train_loss": 13.726996088726253,
    "test_metrics": {
        "mse": 13.684932322692871,
        "mae": 3.5215298175811767,
        "samples": 20000
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
{
    "round": 10,
    "train_loss": 13.727156604630522,
    "test_metrics": {
        "mse": 13.68490050048828,
        "mae": 3.5215259605407714,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
}
],
"final_metrics": {
    "mse": 13.68490050048828,
    "mae": 3.5215259605407714,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
}
}

```

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.736760905458635,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68513099822998,3.5215539653778074
2,13.73491219975885,,,,,13.685111654663086,3.5215516830444336
3,13.74112209662515,,,,,13.685089517211914,3.521549138259888
4,13.740059274786885,,,,,13.685065866088868,3.521546283340454
5,13.731428745961571,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685040350341797,3.5215431350708006
6,13.726476204674901,,,,,13.68501517791748,3.5215401008605958
7,13.731320599997662,,,,,13.684988584899902,3.521536766433716
8,13.72987025365621,,,,,13.684961358642578,3.5215334129333495
9,13.726996088726253,,,,,13.684932322692871,3.5215298175811767
10,13.727156604630522,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68490050048828,3.5215259605407714
FINAL,,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68490050048828,3.5215259605407714

```

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_42.json

```
{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:57:15.878026",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.733752253548612,
      "test_metrics": {
        "mse": 13.685993676757812,
        "mae": 3.5218222717285155,
        "samples": 20000,
        "NDCG@10": 0.062217199771718815,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.004123646725745421,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.734872546745589,
      "test_metrics": {
        "mse": 13.686004968261718,
        "mae": 3.5218238746643067,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 13.737935499853727,
      "test_metrics": {
        "mse": 13.686016731262207,
        "mae": 3.521825635910034,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 13.732044025551165,
      "test_metrics": {
        "mse": 13.686027822875976,
        "mae": 3.5218271327972412,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ]
}
```

```
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
},
{
  "round": 5,
  "train_loss": 13.72877357683148,
  "test_metrics": {
    "mse": 13.686038833618165,
    "mae": 3.5218287628173828,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.735028349771538,
  "test_metrics": {
    "mse": 13.686049507141114,
    "mae": 3.5218302474975585,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.734060697267678,
  "test_metrics": {
    "mse": 13.686059539794922,
    "mae": 3.5218316707611086,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.726528907965097,
  "test_metrics": {
    "mse": 13.686069602966308,
    "mae": 3.5218331199645996,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.732529271134387,
  "test_metrics": {
    "mse": 13.686078648376466,
    "mae": 3.521834460067749,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
```

```

    },
    {
      "round": 10,
      "train_loss": 13.727845996262907,
      "test_metrics": {
        "mse": 13.686086491394043,
        "mae": 3.5218355697631836,
        "samples": 20000,
        "NDCG@10": 0.06462270726815646,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.004123646725745421,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ],
  "final_metrics": {
    "mse": 13.686086491394043,
    "mae": 3.5218355697631836,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  }
}

```

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.733752253548612,0.062217199771718815,0.06,0.006000000000000001,0.004123646725745421,13.685993676757812
,3.5218222717285155
2,13.734872546745589,,,,,13.686004968261718,3.5218238746643067
3,13.737935499853727,,,,,13.686016731262207,3.521825635910034
4,13.732044025551165,,,,,13.686027822875976,3.5218271327972412
5,13.72877357683148,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686038833618165,3
.5218287628173828
6,13.735028349771538,,,,,13.686049507141114,3.5218302474975585
7,13.734060697267678,,,,,13.686059539794922,3.5218316707611086
8,13.726528907965097,,,,,13.686069602966308,3.5218331199645996
9,13.732529271134387,,,,,13.686078648376466,3.521834460067749
10,13.727845996262907,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686086491394043
,3.5218355697631836
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686086491394043,3.521835569763
1836

```

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T03:00:33.121918",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {

```

```
"round": 1,
"train_loss": 13.739092155123583,
"test_metrics": {
  "mse": 13.688507695007324,
  "mae": 3.522353038024902,
  "samples": 20000,
  "NDCG@10": 0.0577968700397596,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 13.732709845652328,
  "test_metrics": {
    "mse": 13.688491870117188,
    "mae": 3.5223504791259765,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 13.730281041256085,
  "test_metrics": {
    "mse": 13.688476957702637,
    "mae": 3.522347919845581,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.733265482019936,
  "test_metrics": {
    "mse": 13.688458750915528,
    "mae": 3.5223448657989502,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.736973903328403,
  "test_metrics": {
    "mse": 13.688440277099609,
    "mae": 3.5223418727874756,
    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
}
```

```
{
  "round": 6,
  "train_loss": 13.737321372608362,
  "test_metrics": {
    "mse": 13.688421134948731,
    "mae": 3.5223387226104736,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.730427924763708,
  "test_metrics": {
    "mse": 13.688401861572265,
    "mae": 3.5223355705261232,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.730882347794372,
  "test_metrics": {
    "mse": 13.68837836151123,
    "mae": 3.522331908798218,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.728986479967666,
  "test_metrics": {
    "mse": 13.688355522155762,
    "mae": 3.5223282726287843,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 13.73102463094966,
  "test_metrics": {
    "mse": 13.688331869506836,
    "mae": 3.522324473571777,
    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 13.688331869506836,
  "mae": 3.522324473571777,
```



```

    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  }
}

```

File: results_backup/dp_inf_alpha_0.1_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.739092155123583,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688507695007324,3.5223530380249
02
2,13.732709845652328,,,,,13.688491870117188,3.5223504791259765
3,13.730281041256085,,,,,13.688476957702637,3.522347919845581
4,13.733265482019936,,,,,13.688458750915528,3.5223448657989502
5,13.736973903328403,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688440277099609,3.5223418727874
756
6,13.737321372608362,,,,,13.688421134948731,3.5223387226104736
7,13.730427924763708,,,,,13.688401861572265,3.5223355705261232
8,13.730882347794372,,,,,13.68837836151123,3.522331908798218
9,13.728986479967666,,,,,13.688355522155762,3.5223282726287843
10,13.73102463094966,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688331869506836,3.5223244735717
77
FINAL,,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688331869506836,3.522324473571777

```

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:09:30.261904",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.72604914243546,
      "test_metrics": {
        "mse": 13.685133767700195,
        "mae": 3.521554398727417,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.721697556470144,
      "test_metrics": {
        "mse": 13.685116149902344,
        "mae": 3.5215523292541504,
        "samples": 20000
      }
    }
  ]
}

```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 13.716468174163426,
    "test_metrics": {
      "mse": 13.685094821166992,
      "mae": 3.521549824142456,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 13.72688887396247,
    "test_metrics": {
      "mse": 13.68507374572754,
      "mae": 3.521547329711914,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 13.7228971265283,
    "test_metrics": {
      "mse": 13.685051594543458,
      "mae": 3.5215446388244627,
      "samples": 20000,
      "NDCG@10": 0.039105890351920955,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.0023614718614718613,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 13.731610071622235,
    "test_metrics": {
      "mse": 13.685030517578125,
      "mae": 3.521542232513428,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 13.714259418429437,
    "test_metrics": {
      "mse": 13.685006246948243,
      "mae": 3.521539258956909,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
}
```

```

    }
  },
  {
    "round": 8,
    "train_loss": 13.722299991741497,
    "test_metrics": {
      "mse": 13.68498218383789,
      "mae": 3.5215363761901854,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 13.716793745279674,
    "test_metrics": {
      "mse": 13.684957339477538,
      "mae": 3.521533363723755,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 13.716583412823606,
    "test_metrics": {
      "mse": 13.684931239318848,
      "mae": 3.5215301780700683,
      "samples": 20000,
      "NDCG@10": 0.039105890351920955,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.0023614718614718613,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 13.684931239318848,
  "mae": 3.5215301780700683,
  "samples": 20000,
  "NDCG@10": 0.039105890351920955,
  "Hit@10": 0.04,
  "Precision@10": 0.004,
  "Recall@10": 0.0023614718614718613,
  "num_users_evaluated": 100
}
}

```

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.72604914243546,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685133767700195,3.521554398727
417
2,13.721697556470144,,,,,13.685116149902344,3.5215523292541504
3,13.716468174163426,,,,,13.685094821166992,3.521549824142456
4,13.72688887396247,,,,,13.68507374572754,3.521547329711914
5,13.7228971265283,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685051594543458,3.5215446388244
627
6,13.731610071622235,,,,,13.685030517578125,3.521542232513428
7,13.714259418429437,,,,,13.685006246948243,3.521539258956909
8,13.722299991741497,,,,,13.68498218383789,3.5215363761901854
9,13.716793745279674,,,,,13.684957339477538,3.521533363723755
10,13.716583412823606,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684931239318848,3.5215301780

```

700683

FINAL,,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684931239318848,3.5215301780700683

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_42.json

```
-----
{
  "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:07:40.304687",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.716136363102116,
      "test_metrics": {
        "mse": 13.685996994018554,
        "mae": 3.5218226345062256,
        "samples": 20000,
        "NDCG@10": 0.06191074357045362,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.004123646725745421,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.72660259513691,
      "test_metrics": {
        "mse": 13.686008929443359,
        "mae": 3.521824362945557,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 13.709877476344737,
      "test_metrics": {
        "mse": 13.686023374938966,
        "mae": 3.521826485824585,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 13.717380628722836,
      "test_metrics": {
        "mse": 13.686035110473632,
```

```
"mae": 3.521828102493286,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 5,
  "train_loss": 13.722983389297813,
  "test_metrics": {
    "mse": 13.686049906921387,
    "mae": 3.521830167007446,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.0060000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.715259929181203,
  "test_metrics": {
    "mse": 13.686062969970703,
    "mae": 3.5218320011138915,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.711035469856222,
  "test_metrics": {
    "mse": 13.686077563476562,
    "mae": 3.5218339515686035,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.717059906595148,
  "test_metrics": {
    "mse": 13.68608907470703,
    "mae": 3.521835577774048,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.726437523744785,
  "test_metrics": {
    "mse": 13.686100819396973,
    "mae": 3.5218372089385985,
    "samples": 20000
  },
  "aggregation": {
```

```

        "num_clients": 100,
        "total_samples": 80000
    },
    {
        "round": 10,
        "train_loss": 13.716294993644347,
        "test_metrics": {
            "mse": 13.686113897705079,
            "mae": 3.5218390003204347,
            "samples": 20000,
            "NDCG@10": 0.06462270726815646,
            "Hit@10": 0.06,
            "Precision@10": 0.006000000000000001,
            "Recall@10": 0.004123646725745421,
            "num_users_evaluated": 100
        },
        "aggregation": {
            "num_clients": 100,
            "total_samples": 80000
        }
    }
],
"final_metrics": {
    "mse": 13.686113897705079,
    "mae": 3.5218390003204347,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
},
"attack_results": {
    "mia": {
        "train_accuracy": 1.0,
        "auc": 0.54805,
        "accuracy": 0.545,
        "true_positive_rate": 0.48,
        "false_positive_rate": 0.39
    },
    "inversion": {
        "top_k_accuracy": 0.0,
        "top_k": 10,
        "num_users_evaluated": 50
    }
}
}

```

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.716136363102116,0.06191074357045362,0.06,0.006000000000000001,0.004123646725745421,13.685996994018554,
3.5218226345062256
2,13.72660259513691,,,,,13.686008929443359,3.521824362945557
3,13.709877476344737,,,,,13.686023374938966,3.521826485824585
4,13.717380628722836,,,,,13.686035110473632,3.521828102493286
5,13.722983389297813,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686049906921387,
3.521830167007446
6,13.715259929181203,,,,,13.686062969970703,3.5218320011138915
7,13.711035469856222,,,,,13.686077563476562,3.5218339515686035
8,13.717059906595148,,,,,13.68608907470703,3.521835577774048
9,13.726437523744785,,,,,13.686100819396973,3.5218372089385985
10,13.716294993644347,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686113897705079
,3.5218390003204347
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686113897705079,3.521839000320
4347

```

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_456.json

```

{
    "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_456",
    "timestamp": "2026-02-11T02:11:17.955196",
    "config": {

```

```
"num_users": 943,
"num_items": 1682,
"embedding_dim": 64,
"num_clients": 100,
"alpha": 0.5,
"dp_epsilon": null,
"use_dp": false,
"dp_sigma": 0.0,
"dp_clip_norm": 1.0,
"num_rounds": 10,
"local_epochs": 3,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 456
},
"rounds": [
  {
    "round": 1,
    "train_loss": 13.724946379484079,
    "test_metrics": {
      "mse": 13.688507598876953,
      "mae": 3.5223532188415527,
      "samples": 20000,
      "NDCG@10": 0.0577968700397596,
      "Hit@10": 0.09,
      "Precision@10": 0.009,
      "Recall@10": 0.0037602713721248985,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 13.726856706559197,
    "test_metrics": {
      "mse": 13.688492994689941,
      "mae": 3.5223507987976075,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 13.72317979983996,
    "test_metrics": {
      "mse": 13.688478712463379,
      "mae": 3.5223483833312987,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 13.717571462923006,
    "test_metrics": {
      "mse": 13.688463998413086,
      "mae": 3.5223459789276124,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
{
```

```
"round": 5,
"train_loss": 13.72191639376251,
"test_metrics": {
  "mse": 13.688447169494628,
  "mae": 3.522343197631836,
  "samples": 20000,
  "NDCG@10": 0.0577968700397596,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 6,
  "train_loss": 13.726362222109655,
  "test_metrics": {
    "mse": 13.688430990600587,
    "mae": 3.5223404735565187,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.720272413013747,
  "test_metrics": {
    "mse": 13.688412133789063,
    "mae": 3.522337431716919,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.720481877012725,
  "test_metrics": {
    "mse": 13.68839178314209,
    "mae": 3.5223342178344725,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.719456072630846,
  "test_metrics": {
    "mse": 13.688371733093263,
    "mae": 3.5223310562133787,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 13.720019055913703,
  "test_metrics": {
    "mse": 13.688349719238282,
```



```

    "mae": 3.5223275356292723,
    "samples": 20000,
    "NDCG@10": 0.057916521733220525,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 13.688349719238282,
  "mae": 3.5223275356292723,
  "samples": 20000,
  "NDCG@10": 0.057916521733220525,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
}
}

```

File: results_backup/dp_inf_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.724946379484079,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688507598876953,3.5223532188415
527
2,13.726856706559197,,,,,13.688492994689941,3.5223507987976075
3,13.72317979983996,,,,,13.688478712463379,3.5223483833312987
4,13.717571462923006,,,,,13.688463998413086,3.5223459789276124
5,13.72191639376251,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688447169494628,3.52234319763183
6
6,13.726362222109655,,,,,13.688430990600587,3.5223404735565187
7,13.720272413013747,,,,,13.688412133789063,3.522337431716919
8,13.720481877012725,,,,,13.68839178314209,3.5223342178344725
9,13.719456072630846,,,,,13.688371733093263,3.5223310562133787
10,13.720019055913703,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.688349719238282,3.5223275356
292723
FINAL,,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.688349719238282,3.5223275356292723

```

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T03:03:53.002176",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.718662279494495,
      "test_metrics": {
        "mse": 13.685132504272461,
        "mae": 3.5215541271209716,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,

```

```
"Hit@10": 0.04,
"Precision@10": 0.004,
"Recall@10": 0.0023614718614718613,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 13.72615446777861,
  "test_metrics": {
    "mse": 13.685114099121094,
    "mae": 3.521552025985718,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 13.720867875590036,
  "test_metrics": {
    "mse": 13.685094969177246,
    "mae": 3.5215497638702393,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.730046955443349,
  "test_metrics": {
    "mse": 13.685073873901366,
    "mae": 3.521547235107422,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.717933644276929,
  "test_metrics": {
    "mse": 13.685054809570312,
    "mae": 3.5215449241638184,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.726125852344104,
  "test_metrics": {
    "mse": 13.685032702636718,
    "mae": 3.521542219543457,
    "samples": 20000
```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 13.716970651184802,
    "test_metrics": {
      "mse": 13.685010177612305,
      "mae": 3.5215395736694335,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 13.714648433277425,
    "test_metrics": {
      "mse": 13.684988346862793,
      "mae": 3.52153685836792,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 13.718525544985368,
    "test_metrics": {
      "mse": 13.68496371307373,
      "mae": 3.5215338943481447,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 13.712659805782865,
    "test_metrics": {
      "mse": 13.684937284851074,
      "mae": 3.5215306037902834,
      "samples": 20000,
      "NDCG@10": 0.039105890351920955,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.0023614718614718613,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 13.684937284851074,
  "mae": 3.5215306037902834,
  "samples": 20000,
  "NDCG@10": 0.039105890351920955,
  "Hit@10": 0.04,
  "Precision@10": 0.004,
  "Recall@10": 0.0023614718614718613,
  "num_users_evaluated": 100
}
```

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_123_summary.csv

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_42.json

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{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T03:02:12.285114",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.721815986879625,
      "test_metrics": {
        "mse": 13.685995581054687,
        "mae": 3.5218225173950195,
        "samples": 20000,
        "NDCG@10": 0.06191074357045362,
        "Hit@10": 0.06,
        "Precision@10": 0.0060000000000000001,
        "Recall@10": 0.004123646725745421,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.73389563101359,
      "test_metrics": {
        "mse": 13.686009576416016,
        "mae": 3.5218245162963866,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ],
  {

```

```
"round": 3,
"train_loss": 13.726155040955048,
"test_metrics": {
  "mse": 13.686022770690919,
  "mae": 3.5218263854980467,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 4,
  "train_loss": 13.717232466348856,
  "test_metrics": {
    "mse": 13.686036561584473,
    "mae": 3.521828374862671,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.717633917105777,
  "test_metrics": {
    "mse": 13.686050863647461,
    "mae": 3.521830379867554,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.73222987273084,
  "test_metrics": {
    "mse": 13.686063023376464,
    "mae": 3.5218321056365967,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.717393401576437,
  "test_metrics": {
    "mse": 13.686075192260741,
    "mae": 3.5218337844848633,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.7209430880483,
  "test_metrics": {
    "mse": 13.686087240600585,
```

```

        "mae": 3.5218354804992678,
        "samples": 20000
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
{
    "round": 9,
    "train_loss": 13.716816745731371,
    "test_metrics": {
        "mse": 13.686100651550293,
        "mae": 3.5218373363494875,
        "samples": 20000
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
{
    "round": 10,
    "train_loss": 13.719377080444453,
    "test_metrics": {
        "mse": 13.686112432861329,
        "mae": 3.521839047241211,
        "samples": 20000,
        "NDCG@10": 0.06462270726815646,
        "Hit@10": 0.06,
        "Precision@10": 0.006000000000000001,
        "Recall@10": 0.004123646725745421,
        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
}
],
"final_metrics": {
    "mse": 13.686112432861329,
    "mae": 3.521839047241211,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
}
}

```

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.721815986879625,0.06191074357045362,0.06,0.006000000000000001,0.004123646725745421,13.685995581054687,
3.5218225173950195
2,13.73389563101359,,,,,13.686009576416016,3.5218245162963866
3,13.726155040955048,,,,,13.686022770690919,3.5218263854980467
4,13.717232466348856,,,,,13.686036561584473,3.521828374862671
5,13.717633917105777,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686050863647461,
3.521830379867554
6,13.73222987273084,,,,,13.686063023376464,3.5218321056365967
7,13.717393401576437,,,,,13.686075192260741,3.5218337844848633
8,13.7209430880483,,,,,13.686087240600585,3.5218354804992678
9,13.716816745731371,,,,,13.686100651550293,3.5218373363494875
10,13.719377080444453,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686112432861329
,3.521839047241211
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686112432861329,3.521839047241
211

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